PS 6 & 7 702

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Problem set 6: Model Selection

To demonstrate the application of subset selection methods

Consider the “Hitters” data. For loading the data you need to install ISLR library. We wish to predict a baseball player’s salary on the basis of various predictors. So first we need to choose an optimal model.

1. Clean the data by removing all rows that have missing value in any variable. You will be working with this clean data.

```
rm(list=ls())
set.seed(123)
library(ISLR)
library(leaps)
h=na.omit(Hitters)
head(h)
```

```
##                    AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits
CHmRun
## -Alan Ashby          315   81     7   24  38    39    14   3449   835
69
## -Alvin Davis         479  130    18   66  72    76     3   1624   457
63
## -Andre Dawson        496  141    20   65  78    37    11   5628  1575
225
## -Andres Galarraga    321   87    10   39  42    30     2    396   101
12
## -Alfredo Griffin     594  169     4   74  51    35    11   4408  1133
19
## -Al Newman           185   37     1   23   8    21     2    214    42
1
##                    CRuns CRBI CWalks League Division PutOuts Assists Errors
## -Alan Ashby          321  414    375      N        W     632      43     10
## -Alvin Davis         224  266    263      A        W     880      82     14
## -Andre Dawson        828  838    354      N        E     200      11      3
## -Andres Galarraga     48   46     33      N        E     805      40      4
## -Alfredo Griffin     501  336    194      A        W     282     421     25
## -Al Newman            30    9     24      N        E      76     127      7
##                    Salary NewLeague
## -Alan Ashby         475.0         N
## -Alvin Davis        480.0         A
## -Andre Dawson       500.0         N
```

```
## -Andres Galarraga   91.5          N
## -Alfredo Griffin   750.0          A
## -Al Newman          70.0          A
```

2. Split the data randomly into train-test keeping 80% as training sample and remaining as test sample.

```
n=nrow(h)
t=sample(1:n,round(0.8*n))
train=h[t,]
test=h[-t,]
attach(train)
```

3. Perform the best subset selection on the training sample. While choosing the best model fit up to a 19 variable model.Choose the best model according to adjusted R^2, BIC, AIC and C_p values. In case the best model differs according to the selection criteria, report the best fitted model according to each selection criterion. In case the best model differs according to the selection criteria, report the best fitted model according to each selection criterion.

```
reg1=regsubsets(Salary~.,data=train,nvmax=19)
reg1s=summary(reg1)
```

Best Subset selection according Aadjusted R^2.

```
c_adjr2=coef(reg1,which.max(reg1s$adjr2));c_adjr2

##  (Intercept)        AtBat         Hits        Walks       CAtBat
CRuns
##  154.1594365   -1.8722264    5.7547106    6.3852096   -0.2092154
1.7712871
##         CRBI       CWalks    DivisionW      PutOuts      Assists
Errors
##    1.1419024   -0.8989348 -121.8633686    0.2013298    0.5363804   -
6.4726568
##   NewLeagueN
##   75.8378982

m1=lm(reformulate(termlabels=c(names(c_adjr2)[-
c(1,9,13)],"Division","NewLeague"),response="Salary"),train)
y_h1=predict(m1)
y_test1=predict(m1,newdata=test)
train_mse1=mean((train$Salary-y_h1)^2)
test_mse1=mean((test$Salary-y_test1)^2)
data.frame("Train MSE"=train_mse1,"Test MSE"=test_mse1)

##   Train.MSE Test.MSE
## 1   88935.8 130066.9
```

Best Subset selection according BIC

```
c_bic=coef(reg1,which.min(reg1s$bic));c_bic

##  (Intercept)        Walks       CAtBat        CHits       CHmRun
DivisionW
##  149.5853608    5.9582020   -0.5241611    2.0687418    1.7652811 -
129.0165012

m2=lm(reformulate(termlabels=c(names(c_bic)[-
c(1,6)],"Division"),response="Salary"),train)
y_h2=predict(m2)
y_test2=predict(m2,newdata=test)
train_mse2=mean((train$Salary-y_h2)^2)
test_mse2=mean((test$Salary-y_test2)^2)
data.frame("Train MSE"=train_mse2,"Test MSE"=test_mse2)

##   Train.MSE Test.MSE
## 1  97026.43 153090.5
```

Best Subset selection according Mallow's C_p measure:

```
c_cp=coef(reg1,which.min(reg1s$cp));c_cp

##  (Intercept)        AtBat         Hits        Walks       CAtBat
CRuns
##  134.1593585   -1.9328699    5.8271814    6.5037245   -0.2063323
1.7820180
##         CRBI       CWalks    DivisionW      PutOuts      Assists
NewLeagueN
##    1.1120229   -0.8879140 -121.2135304    0.1925628    0.3316973
71.1433129

m3=lm(reformulate(termlabels=c(names(c_cp)[-
c(1,9,12)],"Division","NewLeague"),response="Salary"),train)
y_h3=predict(m3)
y_test3=predict(m3,newdata=test)
train_mse3=mean((train$Salary-y_h3)^2)
test_mse3=mean((test$Salary-y_test3)^2)
data.frame("Train MSE"=train_mse3,"Test MSE"=test_mse3)

##   Train.MSE Test.MSE
## 1  89790.63 125652.3
```

4. Find the training MSE and test MSE for each of the best models selected according to different criterion as mentioned in 3 and compare their values.

```
crit=c("Adjusted R sq.","BIC","Mallow's Cp")
Train_MSE_1=c(train_mse1,train_mse2,train_mse3)
Test_MSE_1=c(test_mse1,test_mse2,test_mse3)
data.frame("Criterion"=crit,"Training MSE"=Train_MSE_1,"Testing
MSE"=Test_MSE_1)
```

```
##        Criterion Training.MSE Testing.MSE
## 1 Adjusted R sq.     88935.80    130066.9
## 2            BIC     97026.43    153090.5
## 3    Mallow's Cp     89790.63    125652.3
```

5. Repeat 3-4 using forward stepwise selection method.

```
reg2=regsubsets(Salary~.,data=train,method="forward",nvmax=19)
reg2s=summary(reg2)
c_adjr2=coef(reg2,which.max(reg2s$adjr2))
m1=lm(reformulate(termlabels=c(names(c_adjr2)[-
c(1,11,15)],"Division","NewLeague"),response="Salary"), train)
y_h1=predict(m1)
y_test1=predict(m1,newdata=test)
train_mse1=mean((train$Salary-y_h1)^2)
test_mse1=mean((test$Salary - y_test1)^2)
c_bic=coef(reg2,which.min(reg2s$bic))
m2=lm(reformulate(termlabels=c(names(c_bic)[-
c(1,8)],"Division"),response="Salary"),train)
y_h2=predict(m2)
y_test2=predict(m2,newdata=test)
train_mse2=mean((train$Salary-y_h2)^2)
test_mse2=mean((test$Salary-y_test2)^2)
c_cp=coef(reg2,which.min(reg2s$cp))
m3=lm(reformulate(termlabels=c(names(c_cp)[-
c(1,8,12)],"Division","NewLeague"),response="Salary"),train)
y_h3=predict(m3)
y_test3=predict(m3,newdata=test)
train_mse3=mean((train$Salary-y_h3)^2)
test_mse3=mean((test$Salary-y_test3)^2)
Measure = c("Adjusted R sq.","BIC","Mallow's Cp")
Train_MSE_2 = c(train_mse1,train_mse2,train_mse3)
Test_MSE_2 = c(test_mse1,test_mse2,test_mse3)
data.frame("Criterion"=crit,"Training MSE"=Train_MSE_2,"Testing
MSE"=Test_MSE_2)
```

```
##        Criterion Training.MSE Testing.MSE
## 1 Adjusted R sq.     88287.08    134001.7
## 2            BIC     95658.81    146540.9
## 3    Mallow's Cp     90888.44    145923.9
```

6. Repeat 3-4 using backward stepwise selection method.

```
reg3=regsubsets(Salary~.,data=train,nvmax=19,method="backward")
reg3s=summary(reg3)
c_adjr2=coef(reg3,which.max(reg3s$adjr2))
m1=lm(reformulate(termlabels=c(names(c_adjr2)[-
c(1,10,14)],"Division","NewLeague"),response="Salary"),train)
y_h1=predict(m1)
y_test1=predict(m1,newdata=test)
train_mse1=mean((train$Salary-y_h1)^2)
```

```
test_mse1=mean((test$Salary-y_test1)^2)
c_bic=coef(reg3,which.min(reg3s$bic))
m2=lm(reformulate(termlabels=c(names(c_bic)[-
c(1,6)],"Division"),response="Salary"),train)
y_h2=predict(m2)
y_test2=predict(m2,newdata=test)
train_mse2=mean((train$Salary-y_h2)^2)
test_mse2=mean((test$Salary-y_test2)^2)
c_cp=coef(reg3,which.min(reg3s$cp))
m3=lm(reformulate(termlabels=c(names(c_cp)[-
c(1,10,14)],"Division","NewLeague"),response="Salary"), train)
y_h3=predict(m3)
y_test3=predict(m3,newdata=test)
train_mse3=mean((train$Salary-y_h3)^2)
test_mse3=mean((test$Salary-y_test3)^2)
Train_MSE_3=c(train_mse1,train_mse2,train_mse3)
Test_MSE_3=c(test_mse1,test_mse2,test_mse3)
data.frame("Criterion"=crit,"Training MSE"=Train_MSE_3,"Testing
MSE"=Test_MSE_3)
```

```
##        Criterion Training.MSE Testing.MSE
## 1 Adjusted R sq.     88865.02    134736.7
## 2            BIC     97026.43    153090.5
## 3    Mallow's Cp     88865.02    134736.7
```

7. Compare the results of the three methods in terms of choice of best model, train MSE amd test MSE.

```
Method=rep(c("Best","Forward","Backward"),each= 3)
Criterion=rep(crit,times = 3)
Train_MSE=c(Train_MSE_1,Train_MSE_2,Train_MSE_3)
Test_MSE=c(Test_MSE_1,Test_MSE_2,Test_MSE_3)
data.frame(Method,Criterion,"Training MSE"=Train_MSE,"Testing MSE"=Test_MSE)
```

```
##     Method      Criterion Training.MSE Testing.MSE
## 1     Best Adjusted R sq.     88935.80    130066.9
## 2     Best            BIC     97026.43    153090.5
## 3     Best    Mallow's Cp     89790.63    125652.3
## 4  Forward Adjusted R sq.     88287.08    134001.7
## 5  Forward            BIC     95658.81    146540.9
## 6  Forward    Mallow's Cp     90888.44    145923.9
## 7 Backward Adjusted R sq.     88865.02    134736.7
## 8 Backward            BIC     97026.43    153090.5
## 9 Backward    Mallow's Cp     88865.02    134736.7
```

Comparison of Training MSE:

Adjusted R^2 yields the lowest training MSE across all three selection methods, with the forward selection method achieving the smallest value among them. This suggests that forward selection provides the best fit to the training data compared with the other methods

Comparison of Test MSE

Mallow's C_p performs better on average than the other selection criteria, and among the three methods, the best subset method produces the lowest test MSE

Interpretation

- Forward selection achieves the lowest training MSE, but its comparatively higher test MSE suggests mild over-fitting.
- Backward selection shows a similar pattern, though its performance is slightly weaker than forward selection.
- Best subset selection, particularly with offers the most balanced trade-off between bias and variance

Conclusion

Although best subset selection often generalizes better by examining all possible models, stepwise methods such as forward and backward selection are computationally faster but may miss the optimal variable combination

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Problem Set 7: Shrinkage Methods

We would be working with the Hitters dataset available in the ISLR library. We have to predict the Salary based on all available variables. First, we make sure that we delete all rows with one or more missing values.

1. We split the data set into a training set and a test set randomly, keeping 100 observations in the test set and the remaining in the training set.

```
set.seed(123)
rm(list=ls())
library(glmnet)
library(ISLR)
Hitters=na.omit(Hitters)
t= sample(1:nrow(Hitters), 100)
test=Hitters[t,]
train=Hitters[-t,]
```

2. We train the model using least squares and report the train MSE and test MSE.

```
m=lm(Salary~.,data=train)
y_hat=predict(m,newdata=train)
y_test=predict(m,newdata=test)
train_mse=mean((train$Salary-y_hat)^2)
test_mse=mean((test$Salary-y_test)^2)
data.frame("Train MSE"=train_mse,"Test MSE"=test_mse)
```

```
##   Train.MSE Test.MSE
## 1  105334.2 87803.91
```

3. We fit ridge regression models on the training set, with grid values of λ. We report the following:

(i) the dimension of the coefficient matrix so obtained.

(ii) The l_2 norm of the regression coefficients corresponding to the 50th and 60th choice of λ. We comment on the output.

(iii) Test MSE corresponding to these two choices of λ.

```
lambdas=10^seq(10,-2,length = 100)
x_train=model.matrix(Salary~.,train)[,-1]
x_test=model.matrix(Salary~.,test)[,-1]
ridge_model=glmnet(x_train, train$Salary, alpha = 0,lambda=lambdas)
```

(i) Dimension of Coefficient Matrix

```
dim(coef(ridge_model))

## [1]  20 100
```

(ii) Coefficients for 50th and 60th value of λ respectively

```
coef_50=coef(ridge_model)[,50];coef_50

##   (Intercept)         AtBat          Hits         HmRun          Runs
## 438.509969440   0.029684297   0.110523230   0.560867468   0.178502584
##           RBI         Walks         Years        CAtBat         CHits
##   0.252163513   0.238337160   1.002466622   0.002796464   0.010194612
##        CHmRun         CRuns          CRBI        CWalks       LeagueN
##   0.077846378   0.020089803   0.021957346   0.021866228   0.432252998
##     DivisionW       PutOuts       Assists        Errors    NewLeagueN
##  -6.916280599   0.018294166   0.002998934  -0.082552703   0.741567242

coef_60=coef(ridge_model)[,60];coef_60

##  (Intercept)        AtBat         Hits        HmRun         Runs
RBI
## 105.63996577   0.08310078   0.51246876   2.09635144   0.60993278
1.14623958
##        Walks        Years       CAtBat        CHits       CHmRun
CRuns
##   1.01677160   2.77366963   0.01043635   0.04276159   0.27032634
0.08012067
##         CRBI       CWalks      LeagueN    DivisionW      PutOuts
Assists
##   0.09127173   0.05840151  18.35350369 -62.46410535   0.14784491
0.04002532
##       Errors   NewLeagueN
##  -0.86723778  12.21693330
```

(ii). L2 Norm of Coefficients

```
L2_50=sqrt(sum(coef_50^2));L2_50

## [1] 438.5671

L2_60=sqrt(sum(coef_60^2));L2_60

## [1] 124.7542
```

(iii). Test MSE for 50th and 60th λ

```
ridge_pred_50 = predict(ridge_model, s = lambdas[50], newx = x_test)
ridge_pred_60 = predict(ridge_model, s = lambdas[60], newx = x_test)
Test_mse_50 = mean((test$Salary - ridge_pred_50)^2)
Test_mse_60 = mean((test$Salary - ridge_pred_60)^2)
data.frame("Test MSE for 50th Lambda"=Test_mse_50,"Test MSE for 60th
Lambda"=Test_mse_60)

##   Test.MSE.for.50th.Lambda Test.MSE.for.60th.Lambda
## 1                 167303.1                 97935.56
```

We observe that the test MSE decreases as λ increases.